# ARDMS Vascular Mock Registry Exam C24

Corrected complete version. Questions first, answer key at the end. Feedback removed.

**1.** What console control should be adjusted to improve the quality of the Doppler tracing?

**A.** increase the PW Doppler scale
**B.** reduce the wall filter setting
**C.** raise the baseline
**D.** adjust the placement of the Doppler cursor so it is more parallel to the blood flow

**2.** At what level in the neck does the common carotid artery normally bifurcate?

**A.** At the upper border of the thyroid cartilage
**B.** At the upper level of the thyrocervical trunk
**C.** At the angle of the mandible
**D.** At the lower border of the larynx

**3.** Normal hepatic venous flow will demonstrate:

**A.** two large retrograde diastolic and systolic waves followed by a small retrograde component that corresponds with the atrial contraction
**B.** two large antegrade diastolic and systolic waves followed by a small antegrade component that corresponds with the atrial contraction
**C.** two large antegrade diastolic and systolic waves followed by a small retrograde component that corresponds with the atrial contraction
**D.** two large retrograde diastolic and systolic waves followed by a small antegrade component that corresponds with the atrial contraction

**4.** The ________________________ is described as the segment of the vessel located between the distal ICA and the origin of the anterior communicating artery.

**A.** A2 segment of the ACA
**B.** M1 segment of the MCA
**C.** A1 segment of the ACA
**D.** M3 segment of the MCA

**5.** Which of the following arteries are evaluated during a penile duplex exam for erectile dysfunction?

**A.** ventral
**B.** urethral
**C.** cavernosal
**D.** dorsal

**6.** Which statement is true regarding the image?

**A.** This patient should have an angiogram to confirm the suspected diagnosis of fibromuscular dysplasia.
**B.** There is calcification of the vessels in the vasa vasorum in the arterial wall and the patient should have balloon angioplasty.
**C.** The patient should be evaluated for neointimal hyperplasia.
**D.** There is calcification of the vessels in the vasa vasorum in the arterial wall and the patient should have a TCD exam to evaluate the extent of the disease.

**7.** Which of the following describes a positive result from the Adson maneuver?

**A.** increased hydrostatic pressure
**B.** loss of the radial artery pulse
**C.** pain in the foot with quick dorsiflexion
**D.** loss of pedal pulses

**8.** Vasoconstriction will cause ________________ , while vasodilation will cause ______________.

**A.** increased resistance to flow, decreased resistance to flow
**B.** increased flow volume, decreased flow volume
**C.** increased heart rate, increased blood pressure
**D.** increased flow rate, decreased flow rate

**9.** Which of the following statements regarding a Duplex exam of the upper extremity veins is correct?

**A.** The cephalic vein is easily compressed with light probe pressure, which can be an exam limitation.
**B.** The subclavian and innominate veins normally demonstrate continuous flow.
**C.** Inspiration will cause a decrease in flow within the larger upper extremity veins.
**D.** The internal jugular vein cannot be evaluated by ultrasound due to its course through bony structures of the neck.

**10.** Which of the following normally demonstrates a higher resistance Doppler flow profile?

**A.** external carotid artery
**B.** post-prandial superior mesenteric artery
**C.** renal artery
**D.** hepatic artery

**11.** Which type of aortic aneurysm is associated with an infection of the arterial wall?

**A.** Saccular
**B.** Berry
**C.** Mycotic
**D.** Ectatic

**12.** Normal perforator valves allow blood flow in the perforator to move:

**A.** from the GSV to femoral vein
**B.** from the femoral vein to the GSV
**C.** toward the heart
**D.** toward the ankle

**13.** An aorto-bifemoral graft is used to treat:

**A.** bilateral popliteal stenosis
**B.** unilateral femoral stenosis
**C.** bilateral iliac stenosis
**D.** unilateral iliac stenosis

**14.** When providing a preliminary report for a significant ICA stenosis, what should be included?

**A.** angles of insonation used to obtain Doppler signals
**B.** end systolic velocities in the common and internal carotid arteries
**C.** peak systolic and end diastolic velocities in the common and internal carotid arteries
**D.** hemispheric index for both carotids

**15.** Which of the following is a responsibility of a Sonographer when assisting with a radiofrequency ablation of the great saphenous vein?

**A.** to perform initial puncture for the catheter prior to the vascular surgeon entering the room
**B.** to advance the ultrasound catheter during the procedure
**C.** to send the ultrasound catheter for sterilization after the procedure
**D.** to operate the ultrasound machine controls and/or the probe, as directed by the physician

**16.** Which of the following defines a stroke?

**A.** nerve conduction abnormalities within the cerebral tissue
**B.** acute ischemia in cerebral tissue
**C.** occlusion of the internal carotid artery
**D.** mobile piece of plaque or thrombus

**17.** Longitudinal images are presented of the anterior abdomen just superior to the umbilicus. What is the most common cause of the abnormality seen on the images?

**A.** portal HTN
**B.** polycythemia vera
**C.** malignant HTN
**D.** pulmonary HTN

**18.** Venous flow proximal to an AV fistula becomes ________________ due to the inflow of arterial flow distally

**A.** continuous
**B.** phasic
**C.** pulsatile
**D.** high resistance

**19.** This patient has a brachiocephalic fistula. Use your mouse to position the cursor and click to select the portion of the cephalic vein that is filled with color.

Image-based question.

**20.** The external iliac artery:

**A.** begins at the level of the inguinal ligament
**B.** courses medially through adductor hiatus
**C.** supplies the pelvic organs
**D.** courses along the medial side of the psoas muscle

**21.** What is the Framingham risk score?

**A.** gender specific criteria for estimation of 10-year risk of pulmonary embolism
**B.** criteria used to determine appropriate use of vascular ultrasound
**C.** criteria used to determine appropriate use of physiologic testing
**D.** gender specific criteria for estimation of 10-year risk of cardiovascular disease

**22.** The internal carotid artery terminates into which two branches of the Circle of Willis?

**A.** Anterior and posterior cerebral arteries
**B.** Vertebral and basilar arteries
**C.** middle cerebral and posterior communicator arteries
**D.** middle and anterior cerebral arteries

**23.** What e■ect will the findings on the image have on the Doppler tracings obtained in the iliac veins?

**A.** Both iliac veins will demonstrate continuous flow
**B.** Both iliac veins will demonstrate pulsatile flow
**C.** Both iliac veins will demonstrate phasic flow
**D.** Both iliac veins will demonstrate normal flow patterns due to the extensive venous network in the abdomen that allows for multiple collateral pathways

**24.** A segmental pressure exam is performed for a patient who is experiencing rest pain in the right leg. An ABI of 0.42 was found on the right leg. Which of the following statements is true regarding the findings?

**A.** A PVR exam will provide additional diagnostic information for this patient.
**B.** The patient will be referred to a vascular surgeon for intervention.
**C.** Exercise testing will provide additional diagnostic information for this patient.
**D.** A tourniquet will be applied to the upper thigh and the ankle pressure repeated.

**25.** The most common reason for underestimation of arterial stenosis is:

**A.** improper sample volume angle
**B.** insonation of a vein and artery simultaneously
**C.** improper sample volume location
**D.** insonation of a vein instead of an artery

**26.** The lateral plantar arch artery originates at the ______________________.

**A.** popliteal artery
**B.** peroneal artery
**C.** anterior tibial artery
**D.** posterior tibial artery

**27.** Which of the following can lead to systolic flow reversal in the hepatic veins?

**A.** significant tricuspid regurgitation
**B.** Budd Chiari Syndrome
**C.** IVC thrombus at the level of the renal veins
**D.** portal HTN

**28.** How many cu■s are recommended when performing a bilateral lower extremity segmental pressure evaluation?

**A.** 3
**B.** 5
**C.** 8
**D.** 4

**29.** Which of the following principles produces an equation that relates pressure gradient to flow and resistance within the circulatory system?

**A.** Poiseuille's Law
**B.** Bernoulli Principle
**C.** Doppler Principle
**D.** Raynaud's Phenomenon

**30.** If a patient complains of a recent onset of uncontrollable systemic HTN with average BP readings exceeding 170/90mmHg, what vascular cause should be suspected?

**A.** portal HTN
**B.** subclavian stenosis
**C.** renal artery stenosis
**D.** hepatic congestion

**31.** Which of the following is a characteristic of an ICA waveform with 50-69% stenosis?

**A.** diastolic flow reversal
**B.** decreased diastolic flow with EDV less than 20cm/s
**C.** increased peak systolic and end diastolic velocity
**D.** clear spectral window

**32.** A varicose vein is usually:

**A.** a dilated tributary of the deep system.
**B.** a dilated tributary of the greater saphenous vein.
**C.** a dilated lesser saphenous vein.
**D.** a dilated vessel of the perforator system.

**33.** A patient presents for a segmental pressure exam. She states she has had DVT in her right leg three times over the last 2 years. Today she complains of right leg pain and mild swelling, but no color change. She states that the doctor could not find the pedal pulses on the right side. What should you review in the patient chart prior to starting the exam?

**A.** Medical history for current Coumadin therapy
**B.** Surgical history for endoluminal graft placement
**C.** Lab values for lipid levels
**D.** Lab values for hematocrit levels

**34.** You are evaluating a patient that had a balloon angioplasty on the proximal superficial femoral artery 3 years ago. The acceleration time in the common femoral artery is 230ms. This finding is most suggestive of:

**A.** obstruction in the distal superficial femoral artery
**B.** normal arterial inflow
**C.** iliac artery stenosis
**D.** recurrence of stenosis in the proximal superficial artery

**35.** A patient presents for a bilateral segmental pressure exam and complains of mild leg pain since their heart attack last year. A duplex exam showed medium velocity, triphasic flow with minimal atherosclerosis in the bilateral arteries. The recorded pressures are as follows: Right Arm 110mmHg, Right PTA 88mmHg, Right DPA 92mmHg Left Arm 105mmHg, Left PTA 90mmHg, Left DPA 94mmHg Which of the following could explain the findings?

**A.** The patient most likely has reduced cardiac output causing the right ABI 0.8 and left ABI 0.82.
**B.** The patient most likely has reduced cardiac output causing the right ABI 0.84 and left ABI 0.85.
**C.** The patient is most likely su■ering from Buerger disease.
**D.** The normal arterial duplex exam correlates with the normal lower extremity segmental exam with right ABI 1.20 and left ABI 1.17.

**36.** A patient presents with a script for a "vascular ultrasound" due to a history of Paget Schroetter syndrome. What type of exam should be performed for this patient?

**A.** upper extremity venous Doppler exam
**B.** lower extremity venous Doppler exam
**C.** arterial PPG of the toes
**D.** segmental pressures on the lower extremities

**37.** Pulsus bisferiens in the carotid arteries is a sign of:

**A.** aortic regurgitation
**B.** mitral stenosis
**C.** mitral regurgitation
**D.** pulmonary regurgitation

**38.** The majority of abdominal aortic aneurysms form in what part of the vessel?

**A.** Juxtarenal
**B.** Suprarenal
**C.** Arch
**D.** Infrarenal

**39.** How can you di■erentiate the anterior accessory saphenous vein (AASV) from the great saphenous vein (GSV)?

**A.** the GSV travels in the saphenous compartment but the AASV does not
**B.** look for the alignment sign that is only associated with the AASV
**C.** the AASV courses through the lateral thigh and the GSV courses through the medial thigh
**D.** the AASV travels in the saphenous compartment but the GSV does not

**40.** The superficial venous system of the lower extremities contains vessels that are located

**A.** on the lateral aspect of each leg
**B.** within the connective tissue between the muscle tissue
**C.** superficial to the deep muscular fascia
**D.** adjacent to an artery with the same name

**41.** If there is a left subclavian steal, which brachial artery evaluation will be abnormal?

**A.** Left brachial will have a lower BP and a blunted/monophasic waveform
**B.** Right brachial will have a higher BP and a blunted/monophasic waveform
**C.** Left brachial will have a higher BP and a triphasic waveform
**D.** Right brachial will have a lower BP and a blunted/monophasic waveform

**42.** Which of the following correctly describes the e■ects of walking on the deep venous pressure in the lower extremity in a patient with post phlebitic syndrome?

**A.** Walking causes an immediate, quick drop in pressure in the deep veins at the onset of exercise that quickly returns to pre-exercise levels once the patient stops walking.
**B.** Walking has little to no e■ect on the deep venous pressure, only the arterial pressure is a■ected by exercise.
**C.** Walking causes a significant drop in deep venous pressure that takes several minutes to return to pre-exercise levels once the patient stops walking.
**D.** Walking causes a gradual decrease in pressure during the first 1-2 minutes of exercise that quickly returns to pre-exercise levels once the patient stops walking.

**43.** What measurement is being taken in the image?

**A.** Area stenosis
**B.** Diameter stenosis
**C.** Continuity equation
**D.** IMT

**44.** After an endarterectomy or carotid stent placement, the first follow up exam should be performed:

**A.** immediately following the procedure
**B.** within 5 days of the procedure
**C.** within one month after the procedure
**D.** on the 1-year anniversary date of the procedure

**45.** What are the standard windows used for a complete transcranial Doppler exam?

**A.** suboccipital, transtemporal, transorbital, and submandibular windows
**B.** transtemporal window only, other views performed if abnormalities identified
**C.** suboccipital, transtemporal, and periorbital windows
**D.** suboccipital, transtemporal windows

**46.** Which of the following is not a characteristic of a normal PVR tracing of the lower extremity?

**A.** dicrotic notch
**B.** anacrotic limb
**C.** diastolic flow reversal
**D.** peaked amplitude

**47.** What is an absolute contraindication for sclerotherapy?

**A.** vein location above the inguinal ligament
**B.** patient allergy to injectate
**C.** polycythemia vera
**D.** vein location immediately over a bony prominence

**48.** Which of the following veins originates at the medial wrist and courses up the medial arm to join the axillary vein?

**A.** Radial vein
**B.** Basilic vein
**C.** Ulnar vein
**D.** Cephalic vein

**49.** What is the gold standard method of diagnosing temporal arteritis?

**A.** CTA
**B.** MRA
**C.** biopsy
**D.** duplex ultrasound

**50.** Which of the following describes internalization of the ECA flow?

**A.** The ECA will have a low resistance Doppler flow pattern similar to the waveform in a normal ICA.

**B.** Anatomic variation of the ECA where it is found within the cerebral tissues and connects to the circle of Willis, similar to the distal ICA.

**C.** The left ECA will have a high resistance Doppler flow pattern similar to the ICA.

**D.** The ECA and ICA are transposed anatomically. The ICA has many branches with a smaller caliber and the ECA is larger with no visible branches.

**51.** Which of the following system control adjustments will increase the amount of color Doppler that is superimposed over the 2D image?

**A.** increased wall filter and increased threshold

**B.** increased threshold and increased persistence

**C.** decreased threshold and persistence

**D.** decreased persistence and increased wall filter

**52.** If the distal femoral artery demonstrates a peak velocity of 3.2m/s, the waveform in the posterior tibial artery will most likely be _____________.

**A.** biphasic

**B.** monophasic

**C.** absent

**D.** triphasic

**53.** A patient presents with episodes of left eye amaurosis fugax and right sided paresthesia. These findings are most suggestive of:

**A.** 80% LT ECA stenosis

**B.** RT ICA occlusion

**C.** LT CCA stenosis proximal to current point of evaluation

**D.** LT ICA occlusion

**54.** Which two cursors would be used to measure the velocities used to calculate the resistive index for the flow in this renal artery tracing?

**A.** yellow and red

**B.** blue and yellow

**C.** red and green

**D.** red and blue

**55.** Which of the following patients would benefit from intraoperative TCD monitoring of the left MCA?

**A.** left IJV thrombosis

**B.** left carotid endarterectomy

**C.** right carotid endarterectomy

**D.** right ICA stent placement

**56.** Wernicke aphasia is defined as:

**A.** Inability to speak but can understand others

**B.** Inability to swallow liquids

**C.** Inability to swallow solid foods

**D.** Inability to understand speech, speak, or follow directions

**57.** You are performing a renal Doppler exam. What is the best patient position to visualize the full length of the left renal artery?

**A.** Left lateral decubitus
**B.** Right lateral decubitus
**C.** Supine, contralateral arm above head
**D.** Supine, ipsilateral arm above head

**58.** Which of the following is a normal finding when evaluating a hemodialysis graft?

**A.** Arteriovenous fistula
**B.** Thrombosis
**C.** Pseudoaneurysm
**D.** Aneurysm

**59.** How does congestive heart failure a■ect the carotid artery Doppler waveforms?

**A.** The right CCA waveform will be damped and the left CCA waveform will be increased in velocity and resistance.
**B.** The left CCA waveform will be damped and the right CCA waveform will be increased in velocity and resistance.
**C.** Bilateral CCA waveforms will demonstrate increased resistance and velocity.
**D.** Bilateral CCA waveforms will demonstrate decreased resistance and velocity.

**60.** What is the minimum % area ICA stenosis that is considered hemodynamically significant and treatment is recommended?

**A.** 80%
**B.** 70%
**C.** 75%
**D.** 95%

**61.** The most common site for hepatic artery stenosis in a transplant is ________________________________________.

**A.** at the common hepatic artery bifurcation located just inside the liver near the portal bifurcation
**B.** at the origin of the native hepatic artery from the celiac axis
**C.** at the native hepatic artery anastomosis site located just outside the hilum of the liver
**D.** at the origin of the right and left hepatic arteries

**62.** Transcranial Doppler most commonly uses which Doppler measurement to evaluate flow in the cerebral vessels?

**A.** peak systolic velocity
**B.** resistive index
**C.** time average maximum velocity
**D.** end diastolic velocity

**63.** Which of the following is true regarding the Doppler tracing of the SMA?

**A.** normal SMA flow pattern pre-prandially
**B.** normal SMA flow pattern post-prandially
**C.** SMA stenosis
**D.** There is a pseudoaneurysm of the SMA causing very turbulent flow.

**64.** A patient presents for pre-operative arterial mapping for the TRAM flap breast reconstruction procedure. What vessel(s) should be evaluated?

**A.** epigastric arteries and radial artery
**B.** internal mammary artery and radial artery
**C.** epigastric and internal mammary arteries
**D.** radial artery

**65.** When a patient performs the Valsalva strain, the intra-abdominal pressure ____________________ and the intrathoracic pressure ______________.

**A.** decreases. increases
**B.** increases, decreases
**C.** does not change, increases
**D.** increases, increases

**66.** How can the gastrocnemius veins and the posterior tibial veins be di■erentiated on an ultrasound exam?

**A.** Follow them distally to locate their point of origination
**B.** Document the direction of flow relative to the foot
**C.** Document the direction of flow relative to the knee
**D.** Document the presence of a corresponding artery

**67.** What is the preferred patient position during catheter retraction in an endovenous laser ablation treatment?

**A.** reverse Trendelenburg
**B.** Trendelenburg
**C.** seated with legs dangling
**D.** standing

**68.** A thrill is a normal clinical finding with ________________________, but an abnormal clinical finding with_____________________________.

**A.** a hemodialysis fistula, recent iliac stent placement
**B.** lower extremity vein graft, lower extremity prosthetic graft
**C.** thoracic outlet syndrome, Raynaud syndrome
**D.** recent iliac stent placement, a hemodialysis graft

**69.** Which of the following arteries does not usually have a pair of veins of the same name?

**A.** gastrocnemius artery
**B.** peroneal artery
**C.** femoral artery
**D.** ulnar artery

**70.** A patient presents with left lower extremity pain that awakens him at night. The pain is relieved by sitting at the side of the bed for a few minutes. What would the predicted ABI value be for the left leg?

**A.** 0.8
**B.** 0.6
**C.** 0.4
**D.** above 1.0; these symptoms are most likely unrelated to arterial disease

**71.** A patient presents with a diminished pulse in the right radial artery and a normal pulse in the ulnar artery. These findings are most suggestive of:

**A.** proximal right radial artery stenosis
**B.** innominate artery stenosis
**C.** right brachial artery stenosis
**D.** right subclavian artery stenosis

**72.** When performing a lower extremity venous exam for insu■ciency, you locate an incompetent vein that extends from the ankle to connect to the GSV near the popliteal fossa. Incompetent perforators are also identified in the mid-calf that allow flow between this vein and the GSV. Identify the incompetent vein.

**A.** Small saphenous vein
**B.** Vein of Giacomini
**C.** Posterior arch vein
**D.** Paratibial perforator

**73.** You are performing a TCD exam to assess collateral flow with a right ICA occlusion. You identify flow moving away from the transducer at 90cm/s with decreased pulsatility in the right ophthalmic artery. The left ophthalmic artery demonstrates flow moving toward the transducer with a velocity of 45cm/s. What is indicated by these findings?

**A.** Cross over collateralization
**B.** ECA to ICA collateralization
**C.** Posterior to anterior collateralization
**D.** Normal flow parameters, no collateralization present

**74.** If significant disease is identified on the arterial duplex evaluation, but 4 minutes of exercise has little e■ect on the ankle pressures:

**A.** service should be scheduled for the ultrasound unit
**B.** there are collateral pathways present
**C.** digital pressures must be recorded to diagnose disease in this patient
**D.** post-occlusive reactive hyperemia should be performed

**75.** Which statement is true regarding the image?

**A.** This patient has fibromuscular dysplasia and demonstrates the string of pearls sign on the image.
**B.** There is a stent in the CCA
**C.** There is calcification of the vessels in the vasa vasorum in the arterial wall
**D.** This is a normal CCA image

**76.** Which of the following describes the correct protocol for evaluating suspected popliteal entrapment?

**A.** Use Doppler to evaluate the flow in the distal femoral artery at rest and with plantar extension
**B.** Use Doppler to evaluate the flow in the popliteal vein at rest and with plantar flexion
**C.** Use Doppler to evaluate the flow in the popliteal artery at rest and with plantar flexion
**D.** Evaluate the toe pressures at rest and during plantar extension

**77.** The findings on the image are most suggestive of: view image

**A.** pulmonary hypertension
**B.** aortic valve stenosis
**C.** malignant hypertension
**D.** mitral valve stenosis

**78.** A patient presents for a pre-procedure scan for placement of a central venous catheter in the internal jugular vein. In order to avoid potential hematoma or AV fistula formation, the relationship of the vein to the __________________________ must be documented on today's exam.

**A.** common carotid artery
**B.** external jugular vein
**C.** internal carotid artery
**D.** sternocleidomastoid muscle

**79.** When considering the e■ectiveness of the carotid Doppler exam at your lab, the negative predictive value is 88%, the positive predictive value is 95%, the sensitivity is 85% and the specificity is 90%. Using those parameters, which of the following could be the value for the accuracy of the testing technique?

**A.** 89%
**B.** 84%
**C.** 96%
**D.** 92%

**80.** Which of the following correctly describes an IVC aneurysm?

**A.** Most IVC aneurysms are saccular
**B.** Rarely leads to thrombus formation
**C.** Must be associated with interrupted IVC to be considered a true IVC aneurysm
**D.** Commonly associated with blue toe syndrome

**81.** A patient presents for a carotid Doppler exam due to left arm and leg paresthesia. Which vessel should be closely evaluated for a potential vascular cause of the symptom?

**A.** right ICA
**B.** right vertebral
**C.** left vertebral
**D.** left ICA

**82.** A diabetic patient presents for extremity arterial evaluation with a rest pain bilaterally and blackened tips of the 4th and 5th toe on the left foot. Right Ankle 148mmHg Left Ankle 144mmHg Highest brachial pressure 140mmHg. Which statement is true regarding these findings?

**A.** The signs and symptoms are most likely related to dehydration and poor hygiene.
**B.** An exercise arterial exam is necessary to properly diagnose the patient.
**C.** The signs and symptoms are most likely related to associated venous insu■ciency.
**D.** Toe pressures should be obtained for further evaluation.

**83.** Which of the following are typically characterized as concomitant veins?

**A.** posterior tibial veins
**B.** subclavian vein
**C.** soleal veins
**D.** popliteal vein

**84.** You perform a bilateral venous Doppler due to bilateral swelling and pain in the legs. If both external iliac veins look like the tracing on the image, which of the following should be reported on your technologist worksheet?

**A.** IVC obstruction is suspected
**B.** May-Thurner syndrome is likely
**C.** Severe tricuspid regurgitation is suspected
**D.** Congestive heart failure is likely

**85.** Which of the following is an expected finding with Hypothenar Hammer Syndrome?

**A.** Ulnar artery stenosis or occlusion
**B.** Negative Tinel sign
**C.** Compartment syndrome
**D.** Negative Allen test

**86.** Letter C indicates which of the following structures?

**A.** Internal thoracic artery
**B.** Superficial temporal artery
**C.** Vertebral artery
**D.** Thyrocervical trunk

**87.** The seagull sign refers to which of the following vessels?

**A.** celiac axis
**B.** bilateral renal origins
**C.** three hepatic veins
**D.** common femoral bifurcation

**88.** When reporting spectral Doppler information the Sonographer must:

**A.** classify each waveform as triphasic, biphasic or monophasic
**B.** record the highest quality and highest velocity signals for the final measurements
**C.** document the frequency of the transducer used for the exam should be indicated in the report
**D.** always record the highest velocity obtained from all signals

**89.** Which of the following describes the proper way to measure an abdominal aortic aneurysm?

**A.** outer wall to inner wall
**B.** outer wall to outer wall
**C.** inner wall to inner wall
**D.** inner wall to outer wall

**90.** The subclavian artery becomes the axillary artery:

**A.** when it crosses over the deltoid muscle
**B.** at the level of the first rib
**C.** at the level of the fourth rib
**D.** when it crosses anterior to the clavicle

**91.** The hepatic artery carries _______ of the blood entering the liver.

**A.** 20-30%
**B.** 50-60%
**C.** 80-90%
**D.** 100%

**92.** The Doppler tracing most likely came from which normal lower extremity artery?

**A.** Dorsalis pedis
**B.** Peroneal
**C.** Popliteal
**D.** none of the above

**93.** You are performing an upper extremity venous Doppler exam due to arm swelling and pain. The image displayed is from the left shoulder. Which of the following should be included in your preliminary report?

**A.** There is chronic thrombus formation around an intravenous catheter in the subclavian vein.
**B.** There are echogenic fibrous strands of retracted chronic thrombus present in the subclavian vein.
**C.** The subclavian vein is normal and completely patent with an intravenous catheter in the lumen.
**D.** There is acute thrombus formation around an intravenous catheter in the subclavian vein.

**94.** If there is an acute occlusion of the proximal femoral artery, what will the Doppler waveform look like in the mid femoral artery?

**A.** varies with collateral formation
**B.** blunted
**C.** high resistance
**D.** absent

**95.** When performing post-occlusive reactive hyperemia, pressure measurements in the lower extremities are obtained:

**A.** 1 minute after cu■ release and every 2 minutes until pressures return to baseline levels
**B.** immediately after cu■ release and every 30 seconds until pressures return to baseline levels
**C.** 1 minute after cu■ release and every 30 seconds until pressures return to baseline levels
**D.** immediately after cu■ release and every 2 minutes until pressures return to baseline levels

**96.** A patient that experienced a fall from 15ft yesterday presents with acute hemiparesis and dysphasia. These clinical findings are most suggestive of:

**A.** basilar artery rupture
**B.** polycythemia vera
**C.** carotid dissection
**D.** spontaneous intracerebral aneurysm rupture

**97.** Which vessel courses posterior to the anterior scalene muscle?

**A.** brachial artery
**B.** axillary artery
**C.** aorta
**D.** subclavian artery

**98.** The transducer is placed in the longitudinal position slightly anterior to midline of the medial upper left calf. The beam is angled posterior until the image displayed appears. What vessel is indicated by letter A?

**A.** anterior tibial vein
**B.** peroneal vein
**C.** gastrocnemius vein
**D.** posterior tibial vein

**99.** A patient presents for a carotid Doppler exam due to left sided amaurosis fugax. These findings are most suggestive of stenosis in:

**A.** right vertebral artery
**B.** right ICA
**C.** left ICA
**D.** left vertebral artery

**100.** The carotid siphon:

**A.** is an area of flow separation due to the increase in vessel diameter at the bulb
**B.** supplies blood to the ophthalmic artery
**C.** causes an area of decreased resistance to flow in the distal ICA
**D.** usually produces a waveform with a clear spectral window

**101.** Which of the following parameters is displayed as the y-axis on a Doppler tracing?

**A.** time
**B.** frequency shift
**C.** speed
**D.** motion

**102.** In most patients, the first branch of the ECA is:

**A.** the superficial temporal artery
**B.** the superior thyroidal artery
**C.** the ophthalmic artery
**D.** the internal thoracic artery

**103.** The ability of PW Doppler to determine the ______________________________ is the primary advantage over CW Doppler techniques.

**A.** velocity of flow
**B.** pulsatility of flow
**C.** direction of flow
**D.** location of stenosis in the vessel

**104.** A 75yr old female presents for a transcranial doppler evaluation for suspected basilar artery stenosis. She su■ers from emphysema and cannot lie down for the exam. Which of the following describes how you can evaluate the basilar artery in this patient?

**A.** patient in the left lateral decubitus position, hand supporting the head
**B.** patient seated in chair, lean forward to reach for their toes and flex the neck
**C.** patient seated in chair, neck flexed, and head supported by her arms/hands
**D.** patient seated in chair, neck hyperextended with a wedge behind the head for support

**105.** Which of the following is a direct branch of the aortic arch?

**A.** left vertebral
**B.** left carotid
**C.** right carotid
**D.** right vertebral

**106.** Which intracranial artery is most commonly associated with stroke?

**A.** basilar artery
**B.** posterior cerebral artery
**C.** middle cerebral artery
**D.** anterior cerebral artery

**107.** Lower extremity ulcers are most commonly caused by __________________.

**A.** chronic arterial disease
**B.** chronic venous disease
**C.** acute venous disease
**D.** recurrent cellulitis

**108.** A proximal femoral artery stenosis will cause the __________________ in the mid femoral artery.

**A.** Reynold's number to decrease
**B.** acceleration time to increase
**C.** acceleration time to decrease
**D.** velocity to increase

**109.** The Adson maneuver is helpful in the diagnosis of what vascular disorder?

**A.** Venous insu■ciency
**B.** Thoracic outlet syndrome
**C.** Raynaud syndrome
**D.** Collateral formation with carotid occlusion

**110.** Which of the following is a characteristic of a normal lower extremity arterial PVR waveform?

**A.** prominent dicrotic notch
**B.** low amplitude
**C.** slow downslope
**D.** slow upslope

**111.** Carotid body tumors are more common in:

**A.** women and people living at sea level
**B.** women and people living at high altitudes
**C.** men and people with Buerger disease
**D.** men and people with Ehler-Danlos syndrome

**112.** Amaurosis fugax can be described as:

**A.** TIA of the frontal lobe
**B.** TIA of the eye
**C.** CVA of the eye
**D.** CVA of the frontal lobe

**113.** What term refers to the capability of the vascular beds to alter the resistance to flow to maintain the levels of flow needed for normal function?

**A.** vasoconstriction
**B.** vasodilation
**C.** laminar flow
**D.** autoregulation

**114.** _______________ reflux will be normally seen in the lower extremity superficial system.

**A.** No (0s)
**B.** Less than 0.5s
**C.** Less than 1.5s
**D.** Less than 3s

**115.** What is the proper patient position and respiration for assessing the portal vein diameter?

**A.** left lateral decubitus position with deep inspiration
**B.** right lateral decubitus position with deep inspiration
**C.** supine with inspiration and the Valsalva strain
**D.** supine with quiet respiration

**116.** Which superficial vein listed forms a connection at the elbow between the two main upper extremity superficial veins?

**A.** cephalic vein
**B.** median cubital vein
**C.** radial vein
**D.** basilic vein

**117.** Which of the following factors are related to potential hemodynamic changes within a vessel?

**A.** vessel length and blood viscosity, but not vessel radius
**B.** blood viscosity and vessel radius, but not vessel length
**C.** vessel length and vessel radius, but not blood viscosity
**D.** vessel length, vessel radius and blood viscosity

**118.** The lower extremity segmental pressure exam is most commonly performed with a/an ___________________ to obtain the Doppler signal for the pressure measurements at the ankle.

**A.** 8-10MHz CW Doppler probe
**B.** 2-3MHz CW Doppler probe
**C.** 5-7MHz linear array
**D.** 8-10Mhz vector array

**119.** Which of the following correctly describes a PVR waveform?

**A.** The amplitude of the waveforms from the calf should be higher than the amplitude of the waveforms from the thigh in normal patients
**B.** The patient should be standing for the waveform recording
**C.** The diastolic flow reversal component is exaggerated on a PVR waveform
**D.** Hemodynamic changes due to stenosis will be displayed as a deepening of the dicrotic notch and increased amplitude of the wave

**120.** Which of the following acronyms refers to the standard image file storage format used in medical facilities in the United States?

- **A.** AVI
- **B.** MPEG
- **C.** DICOM
- **D.** JPEG

**121.** If the Doppler cursor angle is underestimated, the velocity of the flow will be ____________________.

- **A.** overestimated
- **B.** underestimated
- **C.** be una■ected
- **D.** dependent solely on the strength of the frequency shift

**122.** If a patient is diagnosed with a pulmonary embolism, what should be the next step in their patient care?

- **A.** V/Q scan
- **B.** lower extremity venous Doppler exam
- **C.** IV Heparin
- **D.** pulmonary angiography

**123.** Which of the following correctly describes Hollenhorst plaques?

- **A.** Heavily calcified plaque that forms in the distal aorta
- **B.** Most patients experience intermittent periods of unilateral blindness
- **C.** Usually seen on optometry exams
- **D.** Heavily calcified plaque that forms in the aortic arch

**124.** A patient presents with a wet ulcer located on the medial calf just above the ankle. What vessel should be evaluated first as the most likely primary cause of the ulcer?

- **A.** posterior tibial artery
- **B.** posterior accessory saphenous vein
- **C.** posterior tibial vein
- **D.** small saphenous vein

**125.** What is the most accurate sonographic indicator for the presence of DVT?

- **A.** flow abnormalities found with PW Doppler
- **B.** loss of augmentation response
- **C.** flow abnormalities found with Color Doppler
- **D.** lack of compressibility

**126.** Which of the following techniques cannot be used to monitor distal flow in the leg while compression therapy for a pseudoaneurysm is performed?

- **A.** PPG sensor on the ankle
- **B.** Pressure monitor on the great toe
- **C.** Doppler evaluation of the PTA or DPA
- **D.** PPG sensor on the great toe

**127.** Which of the following waveforms is most consistent with flow related to a pseudoaneurysm in the groin area?

**A.** A

**B.** C

**C.** D

**D.** B

**128.** When performing a photoplethysmography exam for venous insu■ciency, where should you place the sensor?

**A.** dorsal aspect of the great toe

**B.** groin

**C.** distal pad of the great toe

**D.** medial distal calf

**129.** When venous flow is slow, echogenic particles can be seen near the walls and in the valvular sinus.

**A.** clutter artifact

**B.** spontaneous contrast

**C.** Virchow triad

**D.** misregistration artifact

**130.** Significant valvular disease in the right heart usually leads to:

**A.** cardiac pulsatility in the lower extremity veins

**B.** biphasic flow in the carotid arteries

**C.** tardus parvus waveforms in the carotid arteries

**D.** exaggerated respiratory phasicity in the portal vein

**131.** Priapism is an indication for what type of exam?

**A.** mesenteric duplex

**B.** upper extremity segmental pressure

**C.** carotid duplex

**D.** penile duplex

**132.** A patient presents with dysphagia and hoarseness. The referring physician suspects Ortner syndrome. What type of exam will you be performing?

**A.** transcranial Doppler imaging exam

**B.** upper extremity arterial duplex exam

**C.** transcranial Doppler exam

**D.** bilateral carotid duplex exam

**133.** Which of the following is a contraindication for endovenous laser ablation of the great saphenous vein?

**A.** valvular incompetence in the great saphenous vein

**B.** chronic obstruction of the peroneal veins

**C.** chronic obstruction of the femoral vein

**D.** Baker cyst in the ipsilateral popliteal fossa

**134.** Which of the following describes cerebral cross- over collateralization?

**A.** left ICA flow through posterior communicator to the left vertebral
**B.** flow from the left ACA to the right ACA through the anterior communicating artery
**C.** flow from the right MCA to the left MCA through a cross over collateral vessel
**D.** flow from the left MCA to the right MCA through a cross over collateral vessel

**135.** Which of the following describes the appearance of a waveform distal to a critical carotid stenosis?

**A.** increased diastolic flow
**B.** increased systolic flow
**C.** decreased diastolic flow
**D.** large spectral window

**136.** Which of the following describes the appearance of a significantly abnormal arterial PPG waveform?

**A.** low amplitude with prominent dicrotic notch
**B.** high amplitude with loss of the dicrotic notch
**C.** high amplitude with prominent of the dicrotic notch
**D.** low amplitude with loss of the dicrotic notch

**137.** __________________ uses a balloon tipped catheter to compress the atheroma and expand the vessel lumen.

**A.** An endarterectomy procedure
**B.** An angioplasty procedure
**C.** An atherectomy procedure
**D.** A valvulotome procedure

**138.** You are preparing for a lower extremity segmental pressure exam on a patient with leg pain. She was admitted to the hospital 2 days ago due to extensive lower extremity DVT. How will you proceed?

**A.** check the chart for anticoagulant medication and if it is listed, proceed normally with the exam
**B.** obtain Doppler waveforms and toe-brachial indices
**C.** use cu■s only over areas of the leg where DVT was not identified
**D.** cancel the exam

**139.** Which of the following is a contraindication for US guided compression therapy for a pseudoaneurysm at the groin?

**A.** A pseudoaneurysm that measures 3.2cm in diameter
**B.** Patient that stopped taking Coumadin 3 days ago
**C.** A pseudoaneurysm that has been present for more than 72hrs
**D.** Infection at the puncture site

**140.** A patient presents for a carotid ultrasound due to the presence of bilateral bruits in the proximal neck. What should be the next step in evaluating this patient?

**A.** Evaluate the vascular history section of the patient chart to look for a history of a subclavian steal
**B.** Evaluate the medical history section of the patient chart to look for a history of polycythemia vera
**C.** Evaluate the cardiac history section of the patient chart to look for a history of aortic valve stenosis.
**D.** Evaluate the cardiac history section of the patient chart to look for a history of CHF.

**141.** Which of the following is a system adjustment that is made when changing from a lower extremity arterial exam to a lower extremity venous exam on the same patient?

**A.** change the cursor angle to 30 degrees
**B.** decrease Doppler scale
**C.** decrease transducer frequency
**D.** reverse the color display baseline

**142.** The following pressures were obtained in a patient with left claudication. What are the ABI values for both legs? Right: PTA 128mmHg, DPA, 124mmHg, Calf 130mmHg, Thigh 140mmHg, Brachial 130mmHg Left: PTA 98mmHg, DPA 122mmHg, Calf 126mmHg, Thigh 138mmHg, Brachial 126mmHg

**A.** RT 0.98, LT 0.77
**B.** RT 1.01, LT 0.96
**C.** RT 0.98, LT 0.94
**D.** RT 0.95, LT 0.75

**143.** A patient presents with an order for a lower extremity venous Doppler exam to rule out venous insufficiency. This patient should be evaluated in the _____________ position in order to _______________.

**A.** supine, reduce hydrostatic pressure
**B.** standing, increase hydrostatic pressure
**C.** standing, reduce hydrostatic pressure
**D.** supine, increase hydrostatic pressure

**144.** Using the 3rd waveform, place the first cursor used to begin the resistive index measurement. Use your mouse to position the cursor and click to set your answer.

Image-based question.

**145.** A patient presents with a history of right hip and thigh pain with numbness when he walks. The exercise time varies before the pain starts and it is relieved by sitting down. There are no symptoms in his lower leg. These findings are most suggestive of?

**A.** chronic venous disease
**B.** peripheral arterial disease
**C.** Raynaud's disease
**D.** neurogenic cause

**146.** __________________________________ refers to a collection of mucinous material within the adventitial wall layer of the affected vessel

**A.** Baker cyst
**B.** Mucinous dissection
**C.** Vasa vasorum
**D.** Adventitial cyst

**147.** If a patient suffers from ischemic rest pain in the feet, what maneuver can relieve the pain?

**A.** Standing
**B.** Sitting with legs elevated
**C.** Walking
**D.** Valsalva maneuver

**148.** Pulse volume recording:

**A.** is used to assess arterial insu■ciency and DVT
**B.** will demonstrate waveforms that are nearly identical to the analog Doppler waveforms in a normal patient
**C.** readings should be taken beginning with the distal cu■ and moving proximally
**D.** is commonly performed with a segmental pressure exam

**149.** When reporting spectral Doppler information the Sonographer must:

**A.** always record the highest velocity obtained from all signals
**B.** record the highest quality and highest velocity signals for the final measurements
**C.** classify each waveform as triphasic, biphasic or monophasic
**D.** document the frequency of the transducer used for the exam should be indicated in the report

**150.** You are performing a carotid Doppler exam due to the recent onset of right arm and face paresthesia. There is little plaque present in either carotid system. The right side has normal Doppler flow velocities and waveforms. The attached Doppler images are taken from the proximal left ICA and ECA. Which of the following should be included in your preliminary report?

**A.** There is a probable obstruction of the origin of left carotid artery.
**B.** Congestive heart failure is most likely present.
**C.** There is a probable obstruction of the proximal innominate artery.
**D.** The exam is within normal limits and does not explain the symptoms.

**151.** What is the primary purpose of using a tourniquet during a venous PPG exam of the leg?

**A.** to determine if thrombus is present in the deep or superficial system
**B.** to determine if thrombus is confined to the superficial system
**C.** to determine if venous insu■ciency is confined to the superficial system
**D.** to evaluate the perforators for reflux

**152.** If the systolic brachial pressure is 116mmHg, what is the minimum toe pressure that would be considered normal?

**A.** 80mmHg
**B.** 60mmHg
**C.** 100mmHg
**D.** 70mmHg

**153.** What is the most likely cause of the findings on the image?

**A.** bilateral iliac stenosis
**B.** embolism
**C.** bilateral peroneal artery occlusion
**D.** bilateral popliteal entrapment

**154.** Which of the following describes the sonographic appearance of a transjugular intrahepatic portosystemic shunt?

**A.** anechoic tube, without distinctive wall reflection, which connects the right hepatic vein to the right portal vein
**B.** strongly reflective curved structure connecting the right portal vein and right hepatic vein
**C.** shunts are not easily evaluated Sonographically and CTA is the preferred method for evaluation
**D.** requires color Doppler evaluation to be able to visualize the graft within the liver

**155.** Treadmill exercise is normally performed:

**A.** up to 5 minutes
**B.** up to 12 minutes or until symptoms are intolerable
**C.** until symptoms are intolerable
**D.** until symptoms are intolerable or up to 5 minutes

**156.** A patient presents with a 2-year history of a brachiocephalic hemodialysis fistula. She currently complains that her hand goes numb and turns white when she exercises. You are performing an upper extremity arterial exam and the attached images are from her exam. What will you include in your report?

**A.** There is an obstruction in the palmar arch.
**B.** The proximal radial artery is occluded
**C.** Steal syndrome is present.
**D.** The exam is normal and does not explain the patient's symptoms.

**157.** Which of the following correctly describes sclerotherapy procedures?

**A.** Physicians may inject saline into the deep veins of the extremity in order to increase viscosity and promote the formation of thrombus, which in turn reduces varicosity size.
**B.** Physicians may inject saline into the superficial veins of the extremity in order to cause the vessel to contract and cause fibrosis, which in turn reduces varicosity size.
**C.** Physicians may inject saline into the superficial veins of the extremity in order to cause increased viscosity and promote the formation of thrombus, which in turn reduces varicosity size.
**D.** Physicians may inject saline into the superficial veins of the extremity in order to improve circulation by thinning the blood, which in turn reduces varicosity size.

**158.** What should be done to adjust and improve the color displayed on the attached image?

**A.** Steer color box
**B.** Decrease color baseline
**C.** Increase velocity scale
**D.** Increase PRF

**159.** Which of the following correctly describes how to prepare for an intraoperative vascular ultrasound?

**A.** Cover a high frequency transducer with a sterile sheath that contains gel
**B.** Gather extra packets of sterile gel to be used on the outside of the sterile sheath
**C.** Cover a low frequency transducer with a sterile sheath that contains gel
**D.** Use sterile gloves to open the packaging for supplies used in the procedure

**160.** Which of the following correctly describes the negative predictive value (NPV) of a testing technique?

**A.** If the test result is negative, NPV is the probability that the patient does not have disease
**B.** The ability of a test to rule out disease when it IS NOT present.
**C.** The ability of a test to detect disease when it IS present.
**D.** If the test result is positive, NPV is the probability that the patient actually has the disease

**161.** What is the arrow pointing to on the abdominal angiography exam?

**A.** fibromuscular dysplasia of the left renal artery
**B.** fibromuscular dysplasia of the right renal artery
**C.** normal tortuous splenic artery
**D.** normal right renal artery

**162.** What will you include in your preliminary report for this patient?

**A.** There is a possible obstruction of the left innominate artery.

**B.** There is a possible obstruction of the left innominate artery or the subclavian artery.

**C.** There is a possible obstruction of the right innominate artery or the subclavian artery.

**D.** There is a possible obstruction of the left subclavian artery.

**163.** Which of the following is an important part of Sonographer - Patient communication?

**A.** verifying the patient's name and date of birth

**B.** offering advice on ways to alleviate symptoms until the doctor treats the problem

**C.** explaining the exam and the results

**D.** asking the patient if they are allergic to iodine

**164.** Which of the following describes why there is no aliasing on CW Doppler?

**A.** CW Doppler does not require beam steering to produce a Doppler waveform

**B.** the PW transducer uses a single element and the CW transducer uses two elements

**C.** CW Doppler does not require a sample volume to produce a Doppler waveform

**D.** CW probes are always lower frequency than PW probes

**165.** If a renal artery Doppler exam demonstrates normal flow and normal flow is identified on angiography evaluation, the US results are described as:

**A.** true positive

**B.** false positive

**C.** true negative

**D.** false negative

**166.** Which of the following explains why a patient can receive an US diagnosis of 70% ICA stenosis, but an angiography diagnosis of 50% ICA stenosis?

**A.** Angiography calculates the area stenosis and the US exam offers the diameter stenosis.

**B.** Angiography only evaluates the longitudinal axis of vessel which can lead to over- or under-estimation of the stenosis.

**C.** There was most likely a difference in the patient's cardiac output on the dates the two exams were performed.

**D.** The patient has severe hyperlipidemia with acute disease progression between the two exams.

**167.** Which of the following describes the correct way to measure the resistive index for a renal artery waveform?

**A.** place a caliper at the early systolic peak and the end diastolic velocity

**B.** place a caliper at the onset of the systolic upstroke and at the true systolic peak

**C.** place a caliper at the true systolic peak and the end diastolic velocity

**D.** place a caliper at the onset of the systolic upstroke and at the early systolic peak

**168.** Which artifact is demonstrated on the image?

**A.** side lobes

**B.** reverberation

**C.** cross talk

**D.** volume averaging

**169.** When transmural pressure in a vein is high:

- **A.** the associated artery will distend and flow velocity decreases
- **B.** the associated artery will flatten and flow velocity increases
- **C.** the vein becomes enlarged and rounded
- **D.** the vein becomes flattened and dumbbell shaped

**170.** Which of the following is a correct statement describing the guidelines for wearing a mask?

- **A.** Wash your hands, put on gloves, hold the mask over your mouth by touching the front of the mask. tie the upper strings over your ears toward the top of your head and the bottom strings behind your neck
- **B.** Wash your hands, hold the mask over your mouth by touching the front of the mask. tie the upper strings over your ears toward the top of your head and the bottom strings behind your neck
- **C.** Wash your hands, tie the upper strings over your ears toward the top of your head and the bottom strings behind your neck
- **D.** Wash your hands, put on gloves, tie the upper strings over your ears toward the top of your head and the bottom strings behind your neck

## Answer Key

1. D 2. A 3. C 4. C
5. C 6. C 7. B 8. A
9. A 10. A 11. C 12. A
13. C 14. C 15. D 16. B
17. A 18. C 19. - 20. D
21. D 22. D 23. A 24. B
25. C 26. D 27. A 28. C
29. A 30. C 31. C 32. B
33. A 34. C 35. B 36. A
37. A 38. D 39. B 40. C
41. A 42. A 43. A 44. C
45. A 46. C 47. B 48. B
49. C 50. A 51. B 52. B
53. D 54. B 55. B 56. D
57. B 58. A 59. D 60. B
61. C 62. C 63. C 64. C
65. D 66. A 67. B 68. A
69. C 70. C 71. A 72. C
73. B 74. B 75. - 76. C
77. B 78. A 79. A 80. A
81. A 82. D 83. A 84. A
85. A 86. D 87. A 88. B
89. B 90. B 91. A 92. D
93. D 94. D 95. B 96. C
97. D 98. D 99. C 100. B
101. B 102. B 103. D 104. C
105. B 106. C 107. B 108. B
109. B 110. A 111. B 112. B
113. D 114. B 115. D 116. B
117. D 118. A 119. A 120. C
121. B 122. C 123. C 124. B
125. D 126. A 127. A 128. D
129. - 130. A 131. D 132. B
133. C 134. B 135. C 136. D
137. B 138. B 139. D 140. C
141. B 142. C 143. B 144. -

| | | | |
|---|---|---|---|
| 145. D | 146. D | 147. A | 148. D |
| 149. B | 150. A | 151. C | 152. D |
| 153. - | 154. B | 155. D | 156. C |
| 157. B | 158. A | 159. A | 160. A |
| 161. B | 162. D | 163. A | 164. B |
| 165. C | 166. B | 167. C | 168. C |
| 169. C | 170. C | | |